Attention that eats itself: self-reference flattens the energy landscape of modern Hopfield networks and transformer attention

Nam H. Le

Department of Computer Science and Vermont Complex Systems Institute, University of Vermont, Burlington, VT, 05405, USA

Abstract

Modern Hopfield networks give attention an energy: one retrieval step of a dense associative memory with log-sum-exp energy is exactly the transformer attention operation. Work built on this identity treats the energy landscape as fixed at inference time, yet machine-learning systems increasingly read from stores populated by their own outputs. I show that when a modern Hopfield network, and therefore an attention head, overwrites its stored patterns with its own retrievals, its energy landscape flattens one way. Because every retrieval is a convex combination of stored patterns, some contraction is guaranteed; what I measure is where it ends, how fast it proceeds and how the landscape dies: basins merge in a cascade until a single minimum remains near the data mean, attended uniformly. Pattern separation, the source of exponential capacity, sets only the pace of this flattening, never its endpoint. On twelve pretrained BERT heads, self-reference drives every head to the uniform-attention fixed point, and a head's initial regime predicts how long it resists; the basins disappear before the stored patterns finish travelling. The endpoint coincides with depth-wise rank collapse, reached here along time rather than depth. Energy-based selection over basin widths delays the collapse, and averts it only when the stored patterns encode a problem whose structure retrieval can exploit, in which case the same loop discovers optima it was never shown. Capacity is a clock, not a shield.

Keywords: modern Hopfield networks, dense associative memory,

Email address: nam.le@uvm.edu (Nam H. Le)

transformer attention, energy landscape, model collapse, rank collapse

1. Introduction

Dense associative memories extended the classical Hopfield network (Hopfield, 1982) with sharper interaction functions, buying a storage capacity that grows polynomially (Krotov and Hopfield, 2016) and then exponentially (Demircigil et al., 2017) in dimension. Ramsauer et al. (2021) closed the circle with an exact identity: for the log-sum-exp energy, one step of the retrieval dynamics *is* the attention operation of the transformer (Vaswani et al., 2017),

$$\xi^{\text{new}} \;=\; X\,\text{softmax}\Big(\beta X^\top \xi\Big) \;\;\equiv\;\; z \;=\; V\,\text{softmax}\Big(\tfrac{1}{\sqrt{d}} K^\top q\Big)\,, \qquad (1)$$

with stored patterns X as keys and values, query pattern ξ as the incoming token, and inverse temperature β as $1/\sqrt{d}$ (Fig. 1a). The identity has since become constructive: entire transformer blocks have been derived from a single global energy function (Hoover et al., 2023), energy minimization has been proposed as the substrate of inference-time reasoning (Gladstone et al., 2025), and diffusion models have been recognized as dense associative memories whose spurious minima are the first signatures of generalization (Pham et al., 2025).

All of this work shares one assumption: the landscape stands still. The memories are written by training, and inference only descends the energy they define. But a second literature has been documenting what happens when learning systems consume their own outputs. Generative models retrained on their own samples degrade (Shumailov et al., 2024; Alemohammad et al., 2024), with the severity governed by whether synthetic data replaces or merely accumulates alongside real data (Gerstgrasser et al., 2024; Bertrand et al., 2024); iterated learning in humans converges to the prior of the learner rather than the structure of the data (Griffiths and Kalish, 2007). And a small literature descending from Watson et al. (2011) has studied classical Hopfield networks that relearn their *own attractors*, finding that such "self-modeling" can enlarge the basins of good minima, a line recently reviewed by Froese et al. (2025), entirely within the classical, quadratic-energy setting. Closer to modern Hopfield networks, unlearning schemes that let a network reshape its stored patterns using its own attractors can improve retrieval (Serricchio et al., 2025), and replay selected by Hopfield energy has been proposed to limit forgetting when a memory is retrained (Takeda et al., 2026); in both,

the stored set is shaped by an external procedure rather than overwritten by the network's own retrievals.

A third literature has already found the endpoint I will describe, by a different road. Stacking attention layers drives token representations toward a rank-one configuration in which attention is uniform: pure attention loses rank doubly exponentially with depth (Dong et al., 2021), rank collapse impairs signal propagation and trainability (Noci et al., 2022), the tokens of a deep self-attention flow cluster and eventually merge (Geshkovski et al., 2025), and masks and normalization only change how fast this happens (Wu et al., 2024). In all of these the weights and the context are fixed and the collapse unfolds along *depth*, within a single forward pass. Here nothing is stacked: one head, with its pretrained weights, reads from a memory that it gradually rewrites, so the collapse unfolds in *time*, across rounds of storage. That two different dynamics share the same uniform-attention endpoint is itself part of what I report.

These two literatures have not met. What self-reference does to a *modern* Hopfield network, and therefore, through identity (1), to an attention head, has to my knowledge not been characterized, even though the attention head is precisely the place where modern systems most literally read from stores their own outputs may later populate. Retrieval-augmented systems write generated answers back into the document stores they retrieve from; agents with external memory store their own summaries and reread them; and corpora used to train the next model increasingly contain text produced by the last. In each case a memory that is read by attention is partly written by the reader. This paper makes the characterization, in the field's own vocabulary and partly on the field's own artifacts. My retrieval operator is bit-identical (maximum deviation 2×10^{-14}) to the reference implementation accompanying Ramsauer et al. (2021), and my main experiment runs on pretrained `bert-base-uncased` heads, measured with the same statistic Ramsauer et al. (2021) used to classify real attention heads along the regime axis from *selective retrieval* to *global averaging*.

Part of the answer follows from geometry alone. A retrieval is a convex combination of stored patterns, so a memory that only ever stores its own retrievals can never leave the convex hull of what it started with, and must contract (Proposition 1). The questions geometry leaves open are the ones this paper answers empirically: where inside the hull the memory ends, how fast it gets there, what happens to the basins of attraction on the way, and what it takes to escape.

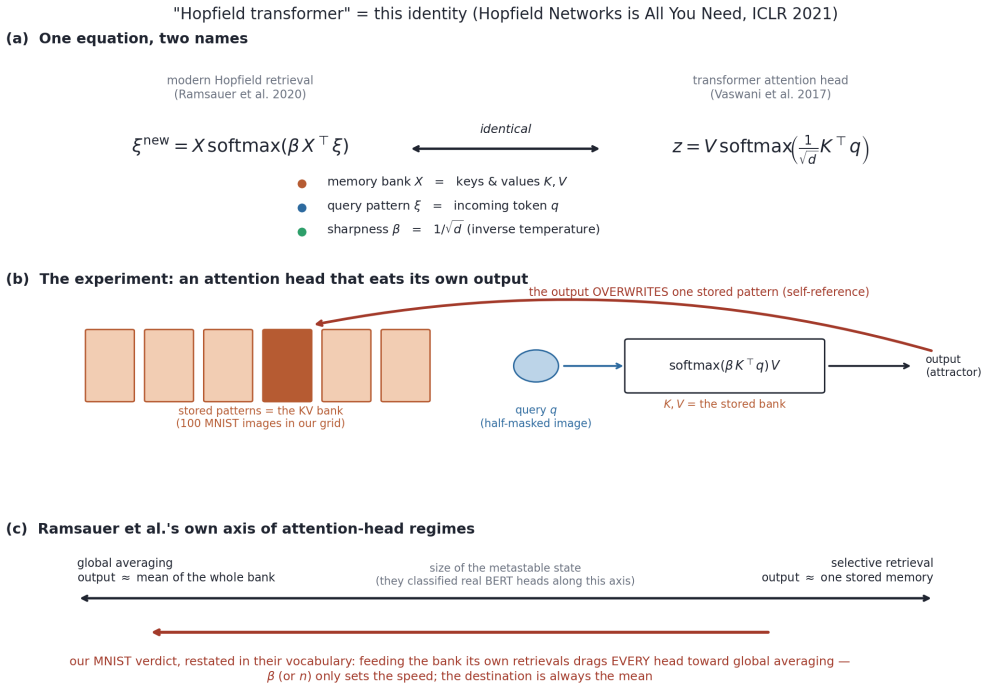


Figure 1: **The object under study.** (a) One retrieval step of a modern Hopfield network is the attention operation (Ramsauer et al., 2021). (b) The self-reference loop: the retrieval output overwrites one stored pattern of the key–value bank; iterating couples the energy landscape to its own minima. (c) Ramsauer et al. (2021) classify attention heads along one axis, from global averaging to selective retrieval, by the size of the metastable state. My result, stated on their axis: self-reference drags every head toward the averaging endpoint, at a speed set by sharpness and starting position.

I find a single organizing picture (Fig. 1b,c). A self-referential attention memory flows one-way along the regime axis to a universal fixed point: a flat landscape with one surviving minimum near the data mean, attended uniformly. Sharpness (Krotov's interaction exponent n, the softmax temperature β, a multiplier τ on pretrained logits) changes only the clock. A positive control shows the escape route: when stored patterns encode a problem with structure to exploit and retrieval is biased so that better memories sit in wider basins, the same loop climbs instead of collapsing, ending beyond everything it was ever shown.

2. Results

2.1. One operation, three substrates

All experiments iterate the same loop. A bank of K stored patterns is queried with a fresh probe; the retrieval output (a softmax- or power-weighted mixture of the stored patterns, Eq. (1)) overwrites one randomly chosen slot of the bank. Nothing else changes: no gradients, no learning rate, no objective. The loop is run on three substrates of increasing realism: (i) a compositional spin-glass world where retrieval quality is externally defined, (ii) MNIST images under a pre-registered protocol, and (iii) the hidden-state banks of twelve pretrained BERT attention heads, retrieved through their own learned projections. Throughout, every judgment of quality or identity is made by an external referee (exact enumeration, a 1-NN classifier on held-out real data, or frozen probe statistics), never by the model's own energy.

The softmax arm of the loop is not an analogy to attention. I certify it against the reference `Hopfield` layer released with Ramsauer et al. (2021): with identity projections and matched scaling, the two implementations agree to a maximum absolute deviation of 2×10^{-14} across random banks and queries. Every result below is therefore a statement about the attention operation itself.

2.2. What the loop guarantees, and what it does not

In the self-referential arms on MNIST and BERT, including the arms with a basin bias, every write is $\xi^{\text{new}} = \sum_i w_i x_i$ with $w_i \geq 0$ and $\sum_i w_i = 1$ (for the power rule, a query with no positive overlap receives uniform weights). This has a consequence that needs no simulation.

Proposition 1. *Let $B_t = \{x_1, \ldots, x_K\}$ be the bank at round t and let each round replace one slot by a convex combination of B_t. Then (i) the convex hull cannot grow, $\operatorname{conv}(B_{t+1}) \subseteq \operatorname{conv}(B_t)$; (ii) no new extreme point is ever created, so an extreme point, once no slot holds it, never returns; and (iii) if all weights are strictly positive, as for a softmax at finite β, the only banks left unchanged by every possible write are those whose patterns all coincide.*

Proof. (i) and (ii): the written pattern lies in $\operatorname{conv}(B_t)$, and removing a point from a set cannot enlarge its hull. (iii): if the bank holds two distinct patterns it has an extreme point x_e; a combination with strictly positive weights over at least two distinct points is not an extreme point of the hull, so overwriting a slot that holds x_e changes the bank. □

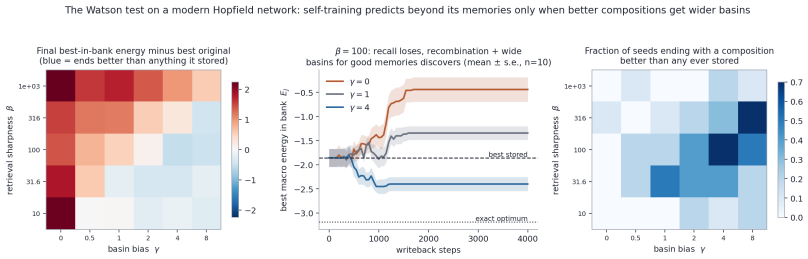


Figure 2: **Self-reference can discover, if quality buys basin width.** Compositional spin-glass substrate. Pure self-training ($\gamma = 0$) is neutral drift: the bank explores novel compositions but fixates, ending farther from the optimum than it began. Biasing retrieval logits by a frozen problem energy ($-\gamma E_J$; better compositions get wider basins, no accept/reject step) reverses the outcome: at moderate sharpness the bank ends strictly better than anything it ever stored.

Self-reference without selection is therefore a consensus process in the sense of DeGroot (1974): it can only lose diversity, and a single repeated pattern is its only resting state. The proposition says nothing, however, about four things, and these are what the experiments measure: (a) *where* in the hull the bank ends (the mean of the data, a class prototype, or an arbitrary stored pattern are all admissible); (b) *how fast* it contracts, and what sets the clock; (c) what happens to the *attractors* of the retrieval map, which are not the stored patterns and can vanish long before the patterns meet; and (d) what it takes to *escape*, which by the proposition requires a write that leaves the hull. On the spin-glass substrate the write is followed by a per-module binarization to the nearest valid composition, a non-convex projection that can produce points outside the hull of the bank; this is the only substrate on which the loop can, even in principle, discover.

2.3. A positive control: when self-reference discovers

To know what collapse is, one must first see its alternative. On the spin-glass substrate, patterns are compositions of module-level configurations of a frustrated macro-problem, so each memory has an externally defined energy E_J and recombining memories can, in principle, assemble optima never stored. Pure self-reference explores (the loop writes dozens of novel valid compositions) but holds nothing: diversity decays monotonically and the bank fixates, ending *worse* than its starting contents (median gap to the global optimum 1.33 $\rightarrow\approx$ 3). This is neutral drift in the population-genetics sense: replacement without selection.

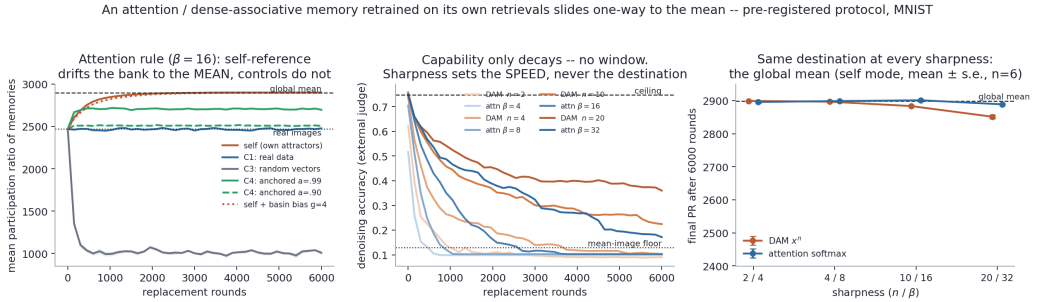


Figure 3: **The drift is to the mean, not to prototypes.** (a) Mean participation ratio (PR) of the memories' correlation profiles under the attention rule: self-reference drives the bank from the real-image level to the global-mean level; replacing memories with fresh real images (C1) does not move it. (b) Denoising accuracy, judged by an external 1-NN classifier, only decays; sharpness (DAM exponent n, attention β) sets the decay rate. (c) Final PR is independent of sharpness for both rules: same destination, different clocks.

The Watson ingredient (Watson et al., 2011), translated from basin *relearning* into its modern form, is to let quality set basin width: retrieval logits are biased by $-\gamma E_J$ with the problem Hamiltonian frozen, so better compositions are easier to fall into, while the dynamics remains pure retrieval. This flips the outcome (Fig. 2). At intermediate sharpness ($\beta \approx 32$–100) and $\gamma \approx 2$–4, the bank ends better than anything it ever stored in 40–70% of seeds (gap $1.33 \to 0.79$; vs. $\gamma = 0$: Mann–Whitney $p = 1.2 \times 10^{-4}$, Cohen's $d = 3.0$; vs. the original bank: Wilcoxon $p = .02$). At $\beta = 1000$ retrieval degenerates into copying, which can preserve but never predict: discovery lives in the recombination regime, not the lookup regime.

2.4. A pre-registered falsification on MNIST

Before the experiments were run, I wrote down a protocol with named falsification criteria. The hypothesis H, motivated by the feature-to-prototype axis of dense associative memories (Krotov and Hopfield, 2016) and by the memorization-to-generalization picture of Pham et al. (2025), predicted that a self-referential memory drifts toward *prototypes* (class-mean-like patterns), with the interaction exponent n setting the speed. A second hypothesis H′ predicted a transient capability window. Both died by their own pre-registered criteria, and the way they died is the finding.

The bank (100 real MNIST images) does drift, and the drift is causally self-referential: feeding fresh real images (control C1) holds the participation-ratio statistic flat (2470→2478), feeding random vectors (C3) drives it in

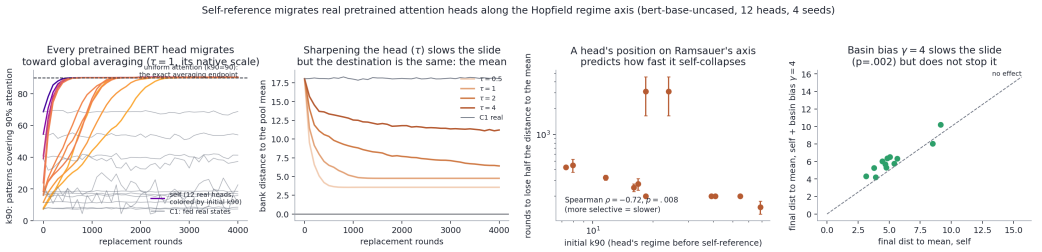


Figure 4: **Twelve pretrained BERT heads under self-reference.** (a) k_{90} (number of stored patterns covering 90% of attention mass, the metastable-size statistic of Ramsauer et al. (2021)) for twelve `bert-base-uncased` heads spanning the regime axis: under self-reference every head converges to $k_{90} = 90.3 \pm 0.1$, the exact value of uniform attention over $K = 100$ patterns; real-fed controls keep their native regimes. (b) Multiplying the pretrained logits by τ changes the speed of the slide, not its direction. (c) Initial regime predicts collapse half-life (Spearman $\rho = -0.72$, $p = .008$). (d) A data-likeness basin bias delays but does not avert the collapse.

the *opposite* direction, and anchoring toward the original bank (C4) slows it monotonically with anchor strength. But the destination is not the class prototypes (PR $\approx$ 2717): it is the global mean of the data (PR $\approx$ 2899), reached at every n and every β (Fig. 3a,c). Capability never peaks; it only decays from the starting ceiling, faster for blunt rules (final denoising accuracy 0.09 at $n = 2$ vs. 0.36 at $n = 20$ after 6,000 rounds; Fig. 3b). The DAM power rule and the attention softmax rule are empirically interchangeable throughout, a dynamical confirmation of identity (1). And the basin bias that rescued the spin glass does *not* rescue MNIST: a mean-blurred image still scores high data-likeness, so selection by similarity has nothing to push against. Where the spin glass had structure to exploit, an at-ceiling representation task has only entropy to lose.

2.5. Real pretrained attention heads migrate to the uniform-attention fixed point

The decisive test runs on the field's own artifact. From `bert-base-uncased` I select twelve heads (at layers 1, 5, 8 and 11, the most averaging, the median, and the most selective head of each by initial k_{90}, spanning $k_{90} \in [5, 68]$ on a bank of $K = 100$ real token hidden states) and iterate the loop through each head's own pretrained W_Q, W_K (values taken as the patterns themselves, so retrieval stays in pattern space). The outcome is universal (Fig. 4). Every bank fixates onto a *single* repeated pattern near the pool mean (distinct slots $\to 1$;

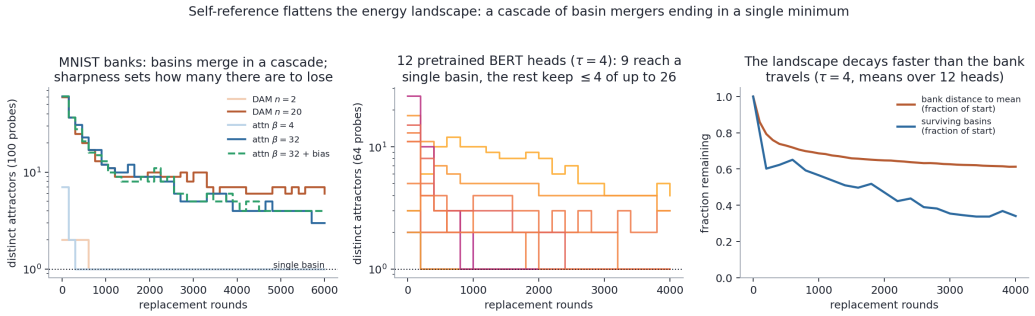


Figure 5: **Flattening, measured directly.** At fixed times the bank is frozen and the retrieval map is iterated to convergence from a fixed set of real probes; distinct limit points count the surviving basins. (a) MNIST banks: sharp rules start with ∼60 basins and lose them in a cascade; blunt rules have almost none to lose. The energy-biased run (dashed) tracks the unbiased cascade. (b) The same census on pretrained BERT retrieval maps ($\tau = 4$): starting with up to 26 basins, nine of twelve heads end with one, the rest with ≤ 4. (c) Basins disappear faster than the bank approaches the mean: the landscape loses its structure while the patterns are still travelling.

effective rank 55 $\to\approx$ 1.5), and the probe statistic lands on $k_{90} = 90.3 \pm 0.1$ across all twelve heads, exactly the value of a uniform softmax over 100 patterns, i.e., the far averaging endpoint of the regime axis. Controls fed real hidden states keep their native diversity of regimes ($k_{90} = 26 \pm 19$; Wilcoxon over heads, $p = 4.9 \times 10^{-4}$). Sharpening the pretrained logits (τ up to 4) slows the slide without changing its direction, and a head's initial position on the axis predicts its collapse half-life (Spearman $\rho = -0.72$, $p = .008$): selectivity is measured in rounds of survival, not in immunity. The basin bias again delays ($p = .002$) but does not save: hidden-state banks, like MNIST, offer similarity-based selection nothing to exploit.

2.6. The landscape flattens by a cascade of basin mergers

The statements above are consequences of a single geometric event, which I measure directly: the number of attractors of the retrieval map, counted by iterating the frozen map to convergence from fixed real probes and deduplicating the limit points. Sharp rules begin with many basins (∼60 of 100 on MNIST at $\beta = 32$ or $n = 20$; up to 26 on sharpened BERT heads) and lose them in a cascade of mergers; blunt rules begin already nearly flat (Fig. 5a,b). The cascade outruns the patterns themselves: at $\tau = 4$ the surviving-basin fraction falls to a third while the bank has covered barely 40% of its distance to the mean (Fig. 5c). The energy landscape dies first; the visible drift of

the patterns is the afterglow. This is the precise sense in which "collapse" is the right word: not a loss of data, but a loss of minima, the heat death of a self-referential energy landscape.

2.7. Three laws

Across substrates (synthetic compositions, images, and pretrained transformer hidden states) the same three regularities hold.

Universal destination. Unselected self-reference has one fixed point. That it has a single one is guaranteed by Proposition 1; where it lies is not, and empirically it is a single basin at (or drifting toward) the data mean, with uniform attention. Prototype formation, the natural first guess, is falsified pre-registeredly: averaging across retrievals beats averaging within classes.

Capacity is a clock, not a shield. Interaction sharpness (n, β, τ) and initial regime position set every timescale in the problem (exploration before fixation, capability half-life, basin survival) and none of the destinations.

Selection needs structure. Basin-width selection converts the loop from drift to discovery exactly when stored patterns encode a problem whose recombinations can be externally better (spin glass), and merely slows the clock when they do not (MNIST, BERT). By Proposition 1 the two ingredients are distinct: a write that can leave the hull (here, the binarizing projection) supplies novelty, and selection decides which novelty is kept. Bias alone, applied to convex writes, can only choose the order in which diversity is lost. The boundary between self-training that collapses and self-training that discovers is a property of the task's structure, not of the architecture's capacity.

3. Discussion

An attractor account of model collapse. The collapse of generative models trained on their own outputs (Shumailov et al., 2024; Alemohammad et al., 2024) is usually described distributionally, as the loss of tails. Restricted to the one module where the identity (1) is exact, I can replace the description with a mechanism: self-reference couples an energy landscape to its own minima, and the coupled system has a Lyapunov-like one-way flow to the flat landscape. The known mitigations map cleanly: accumulating rather than replacing data (Gerstgrasser et al., 2024) bounds the fraction of self-written patterns; verification-based filtering is selection with an external referee; and my basin-bias result sharpens the requirement: similarity to real data is

not a sufficient referee, because the mean is maximally similar to everything. Likewise the convergence-to-the-prior of iterated learning (Griffiths and Kalish, 2007) appears here in its most stripped-down form: the "prior" of a retrieval map is its uniform-attention fixed point.

Relation to rank collapse. Depth-wise rank collapse (Dong et al., 2021; Noci et al., 2022; Geshkovski et al., 2025) and the self-referential collapse reported here end at the same fixed point, uniform attention over a single averaged pattern, but they differ in what moves. In rank collapse the stored keys and values are recomputed at every layer from the tokens of one input, and the collapse is a property of composing attention maps. In self-reference a single map is applied repeatedly while its memory is overwritten, and the collapse is a property of the memory, which persists across inputs. Three observations follow. First, the remedies differ: residual connections and normalization act on the forward pass (Noci et al., 2022; Wu et al., 2024) and have no counterpart in a store that is rewritten in place; what slows my loop is sharpness, initial regime and an external referee. Second, the clock is set by the same quantity in both: sharper attention resists longer, whether measured in layers or in rounds. Third, the basin census adds a statement the depth-wise analyses do not make in energy terms: the attractors of the retrieval map vanish before the patterns finish converging.

For energy-based transformers. Architectures that derive transformer blocks from energies (Hoover et al., 2023) or spend inference compute descending them (Gladstone et al., 2025) inherit an assumption of landscape stationarity. My results are a first map of what happens when that assumption is dropped at the memory level, as it is dropped in practice wherever a retrieval store, cache, or training corpus is populated by the system that reads it. The regime axis of Ramsauer et al. (2021) turns out to be not just a taxonomy but a flow line, and position on it a survival time.

When self-reference is worth having. The spin-glass control recovers, in modern form, the self-modeling effect of Watson et al. (2011) and the conditions of the self-optimization literature (Froese et al., 2025): self-reference adds value precisely when (i) patterns are compositional, so recombination can produce something new, and (ii) an external quality signal is allowed to shape basin geometry. Under those two conditions the loop is a generate-and-test optimizer whose generator is the memory itself; without them it is neutral drift with a guaranteed flat endpoint.

Limitations. The contraction itself is a geometric certainty for convex writes, so the contribution lies in its endpoint, its clock and its basin dynamics

rather than in the fact of collapse. These are small-scale, mechanism-level experiments: one pretrained model, banks of 100 patterns, retrieval-space dynamics rather than full text generation, and BERT's asymmetric $W_Q W_K^\top$ means the head's iterated map is not literally gradient descent on a symmetric energy (I therefore count attractors of the map, which is the operationally relevant object). The pre-registered MNIST protocol also taught me that my drift statistic cannot distinguish prototype formation from mean formation in advance; only the controls made the verdict unambiguous. Scaling the loop to generative language models, where the bank is a KV cache or a corpus and the probe distribution itself shifts, is the natural next experiment; the three laws above are its pre-registered predictions.

4. Methods

Retrieval rules. Power rule (DAM): weights $w_i \propto \max(\cos(q, x_i), 0)^{n-1}$ (Krotov and Hopfield, 2016); softmax rule (attention): $w_i \propto \exp(\beta \cos(q, x_i))$ (Ramsauer et al., 2021); both applied to centred patterns, output $\mu + \sum_i w_i(x_i - \mu)$. On BERT heads the logits are $\tau\,(qW_Q)(x_i W_K)^\top/\sqrt{64}$ with pretrained weights and biases. Certification against the reference implementation of Ramsauer et al. (2021) used identity projections, static patterns, and matched scaling.

Spin-glass substrate. Patterns are concatenations of 12 module configurations (10 alleles each) scoring against a frustrated coupling matrix; the global optimum is known by exact enumeration. Self-training: random valid query $\to$ retrieval $\to$ per-module binarization to the nearest valid composition $\to$ overwrite a random slot with a δ mixture. Basin bias adds $-\gamma E_J(x_i)$ to the logits with J frozen. Grids: $\beta \in 10^{0..3}$ (13 values) $\times$ δ (8 values) $\times$ 6 seeds; selection arm $\beta \in \{10, \ldots, 1000\} \times \gamma \in \{0, \ldots, 8\} \times 10$ seeds.

MNIST protocol. Pre-registered before any run: the participation ratio $\mathrm{PR}(\xi) = (\sum_i c_i^+)^2 / \sum_i (c_i^+)^2$ of a memory's positive correlation profile against 3,000 real images, validated in advance on known cases (real images 2462 $<$ class means 2717 $<$ global mean 2899; random 1030); a difficulty check establishing that the inpainting task needs memory rank 14–18 (ceiling 0.748, mean-image floor 0.130); then a $2 \times 4 \times 6 \times 6$ grid (rule $\times$ sharpness $\times$ mode $\times$ seed), $K = 100$, 6,000 replacement rounds, with modes self, real-fed (C1), random-fed (C3), anchored (C4, $\alpha \in \{0.99, 0.90\}$) and an exploratory data-likeness bias ($\gamma = 4$). Capability judged by masked-pixel reconstruction scored by a 1-NN classifier on held-out real images. Falsification criteria

(monotone PR decrease toward prototypes for H; non-monotone capability peak for H′) were written in advance; both criteria fired negative, as reported.

BERT experiment. `bert-base-uncased`; token hidden states from two public-domain books cached at layers 1, 5, 8, 11 (20,480 states per layer); per layer the most-averaging, median, and most-selective heads by initial k_{90} on a $K = 100$ bank. Dynamics: 4,000 rounds $\times$ $\tau \in \{0.5, 1, 2, 4\}$ $\times$ modes {self, real, random, bias} $\times$ 4 seeds. Frozen probe set of 256 states; k_{90}, attention entropy, effective rank, distance to pool mean, nearest-neighbour distance to real states, and distinct-slot count recorded every 100 rounds at the head's native scale.

Basin census. At each stored time the bank is frozen and the retrieval map iterated to convergence (tolerance 10^{-9} MNIST / 10^{-5} BERT, ≤ 300 iterations) from fixed real probes (100 MNIST / 64 BERT); limit points are de-duplicated at cosine 0.999.

Statistics. Per-seed units throughout; Mann–Whitney U and Cohen's d for independent comparisons, Wilcoxon signed-rank for paired ones, Spearman rank correlation for the half-life law. No statistic in the abstract; all numbers in the text are means over seeds unless stated.

CRediT authorship contribution statement

Nam H. Le: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools in order to draft and edit the text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability

All code, the stored results behind every figure, and the pre-registered MNIST protocol are openly available at `https://github.com/drnamlabs-gif/self-referential-attention`.

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